STOMACH

Regular anatomy notes with enlarged source-PDF diagrams and exam-oriented explanations

Core identity

The stomach is the widest and most distensible part of the alimentary canal between the esophagus and the duodenum.

- Synonyms: gaster in Greek; venter in Latin.
- Main position: upper left abdomen, chiefly under cover of the left costal margin and lower ribs.
- Usual shape: J-shaped, but shape and position vary with build, posture, contents and respiration.

Main functions

The stomach acts as a reservoir, mixer and regulator of food entry into the small intestine.

- Stores food temporarily.
- Mixes food with gastric secretions to form chyme.
- Controls the rate of emptying into the duodenum.
- HCl destroys bacteria; intrinsic factor helps vitamin B12 absorption.

Fig. 7.1 - Location of the stomach

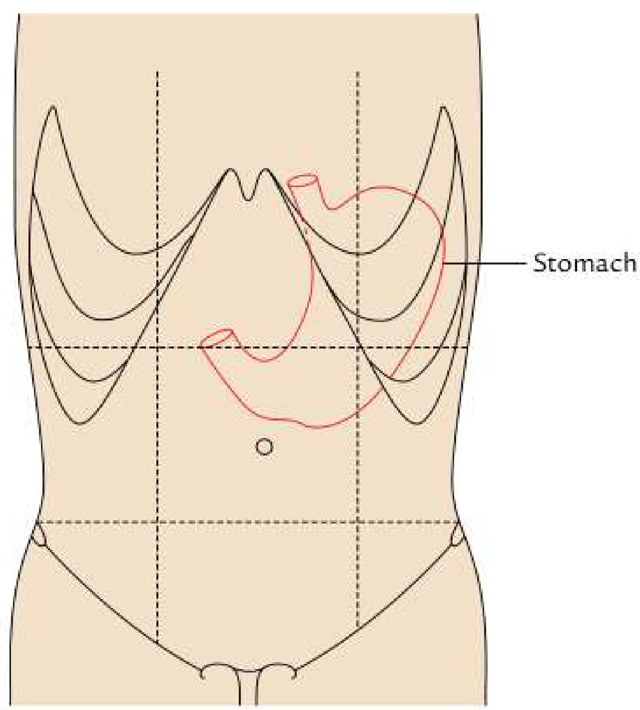


Fig. 7.1 Location of the stomach (demarcated by red line).

The red outline shows the stomach lying mainly in the left hypochondriac and epigastric regions, with extension towards the umbilical region when distended. This image is useful for remembering why epigastric pain, left hypochondrial fullness and left subcostal percussion findings are clinically relevant.

LOCATION, SHAPE, SIZE AND CAPACITY

Location

The stomach is situated in the upper left part of the abdomen. It occupies the left hypochondriac, epigastric and umbilical regions.

- It extends obliquely from the left hypochondriac region into the epigastric region.
- Most of it lies under cover of the left costal margin and lower ribs.
- The cardiac end is near T11, about 2.5 cm to the left of the median plane.

Shape and capacity

The stomach is mostly J-shaped; its long axis passes downward, forward, rightward and finally backward/slightly upward.

- Length: about 10 inches.
- Capacity at birth: about 30 ml.
- Capacity at puberty: about 1000 ml.
- Adult capacity: about 1500-2000 ml.

Fig. 7.2 - Radiological shapes of the stomach

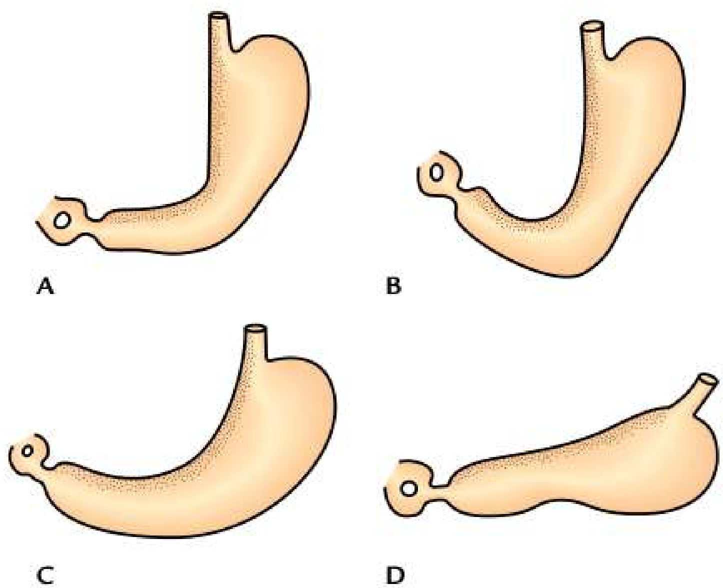


Fig. 7.2 "Types" of shape of the stomach as seen in barium meal X-ray of abdomen: A, reversed L-shaped; B, J-shaped; C, semilunar shaped; D, steer-horn shaped.

Barium meal radiographs may show reversed L-shaped, J-shaped, semilunar or steer-horn forms. A short obese person usually has a high transverse stomach, while a tall thin person commonly has a low elongated stomach. Therefore, the stomach is a mobile organ whose position changes with posture, contents and respiration.

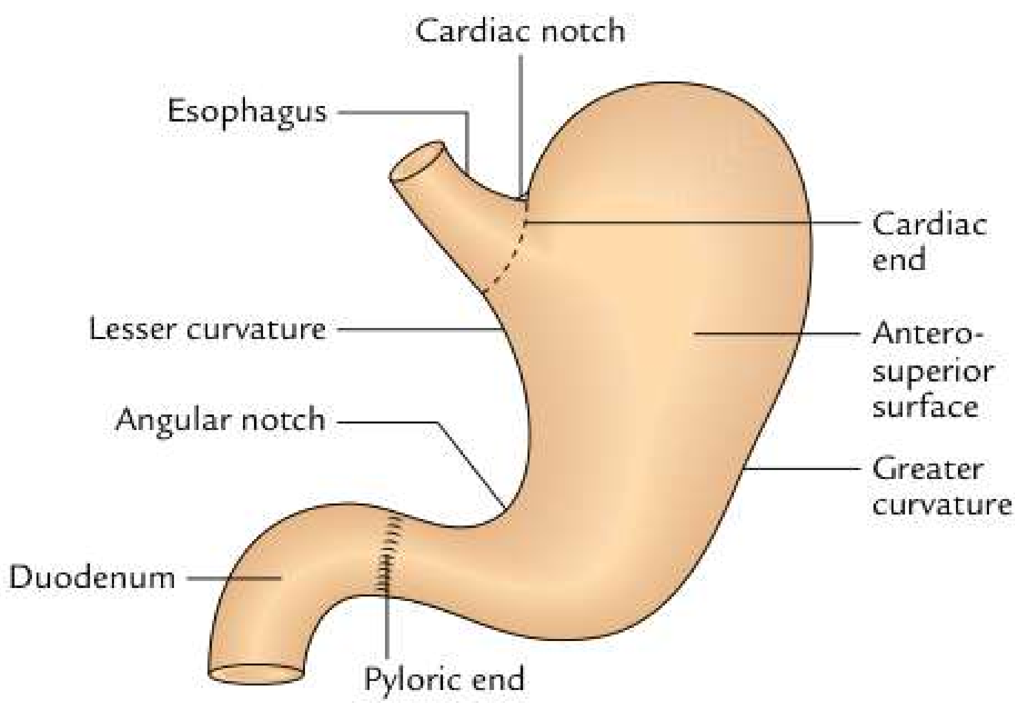
EXTERNAL FEATURES

External features at a glance

For description, the stomach presents two ends, two curvatures and two surfaces. These features form the base for understanding attachments, relations and vascular supply.

- Ends: cardiac end and pyloric end.
- Curvatures: lesser curvature and greater curvature.
- Surfaces: anterior/anterosuperior surface and posterior/posteroinferior surface.
- Important notches: cardiac notch and angular notch/incisura angularis.

Fig. 7.3 - External features of the stomach



The lesser curvature forms the shorter right border and carries the angular notch. The greater curvature forms the longer left border and gives attachment to the greater omentum, gastrosplenic ligament and gastrophrenic ligament. The diagram also shows why the pyloric end is more variable in position than the cardiac end.

Feature	Cardiac orifice	Pyloric orifice
Location	Junction between lower end of esophagus and stomach.	Junction between pyloric end of stomach and first part of duodenum.
Surface level	Behind left 7th costal cartilage, 2.5 cm from sternum; lower border of T11.	About 1.5 cm right of median plane at lower border of L1; transpyloric plane.
Sphincter	No anatomical sphincter; guarded by physiological sphincter.	Has anatomical pyloric sphincter due to thickened circular muscle coat.

PARTS OF THE STOMACH

Four anatomical parts

The stomach is divided into cardiac part, fundus, body and pyloric part.

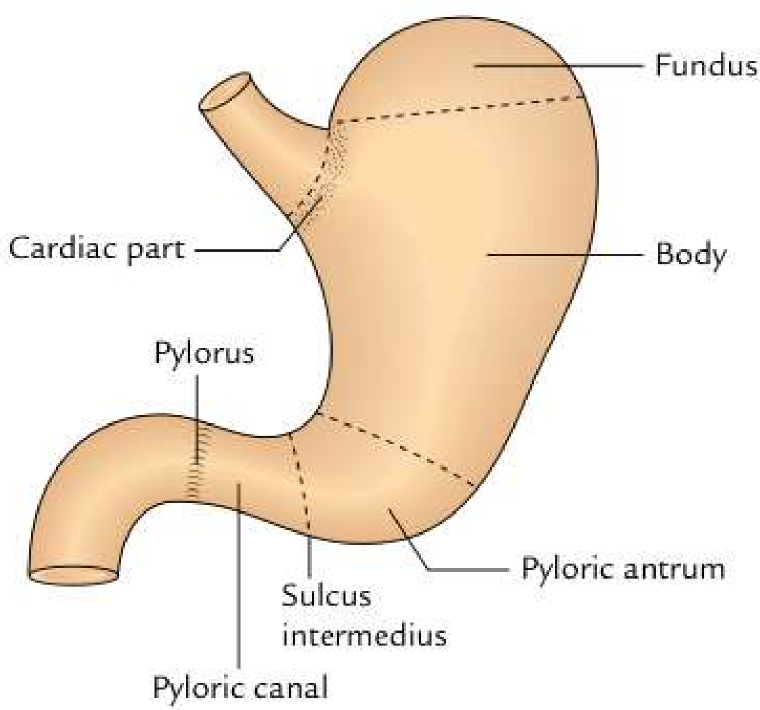
- Cardiac part: around the cardiac orifice.
- Fundus: dome above the horizontal plane of cardiac notch; commonly contains gas.
- Body: largest part, between fundus and pyloric antrum.
- Pyloric part: funnel-shaped outflow region.

Pyloric part

The pyloric part extends from the angular notch to the gastroduodenal junction and is divided into pyloric antrum, pyloric canal and pylorus.

- Pyloric antrum: wider proximal part.
- Pyloric canal: narrow distal tubular part.
- Pylorus: terminal sphincteric region controlling gastric emptying.

Fig. 7.4 - Parts of the stomach



The fundus, body and pyloric region are not just descriptive parts: they also relate to clinical sites of pain, gastric emptying and ulcer/cancer distribution. The sulcus intermedius helps separate pyloric antrum from pyloric canal, while the pylorus marks the muscular gate to the duodenum.

Traube space and fundus

The fundus usually contains gas and appears as a radiolucent shadow under the left dome of the diaphragm. Traube space overlies the fundus and is tympanic on percussion. It is bounded above by the lower border of the left lung, below by the left costal margin, on the left by the spleen and on the right by the left lobe of the liver.

CURVATURES, SURFACES AND PERITONEAL RELATIONS

Curvatures

Lesser curvature is concave, forms the shorter right border and gives attachment to the lesser omentum. Its most dependent point is the angular notch.

Greater curvature is convex, forms the longer left border and gives attachment to greater omentum, gastrosplenic ligament and gastrophrenic ligament.

Surfaces

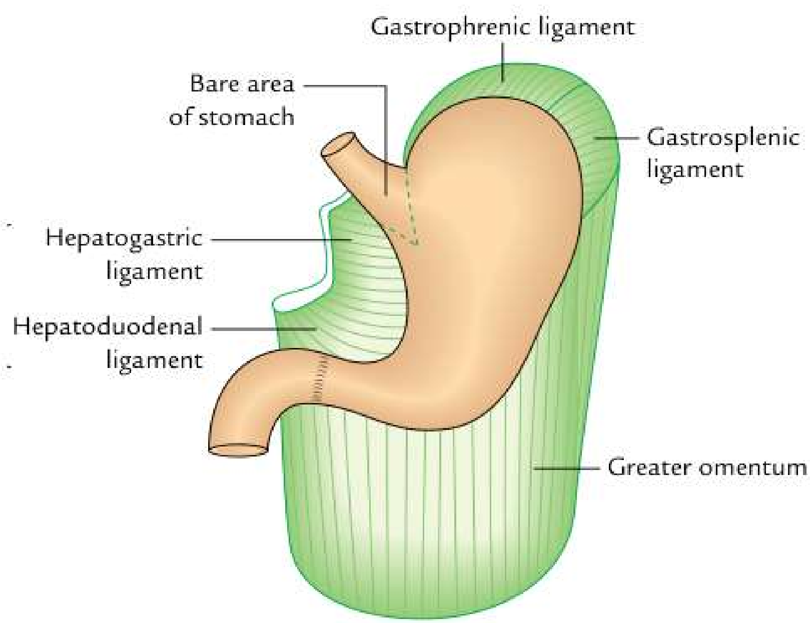
Anterior surface faces forwards and upwards. It is related to liver, diaphragm and anterior abdominal wall.

Posterior surface faces backwards and downwards. Its relations form the stomach bed and are separated from stomach by the lesser sac, except the spleen.

Attachment map of the stomach

For exam answers, connect each border with its peritoneal attachment: the lesser curvature is linked to the liver by the lesser omentum, while the greater curvature is linked to the transverse colon, spleen and diaphragm through the greater omentum, gastrosplenic ligament and gastrophrenic ligament. This is why vessels, lymphatics and surgical planes are described along the curvatures.

Fig. 7.5 - Peritoneal relations of the stomach



This figure summarizes the peritoneal folds attached to the stomach. Lesser omentum attaches to the lesser curvature. Greater omentum descends from the greater curvature. Gastrosplenic ligament passes from fundus/greater curvature to spleen, and gastrophrenic ligament passes from fundus to diaphragm. The bare area near the cardiac region is not covered by peritoneum.

Peritoneal covering

The stomach is covered by peritoneum except where blood vessels run along its curvatures and a small posterior bare area near the cardiac orifice. Peritoneal relations are important because they connect the stomach to liver, spleen, diaphragm and transverse colon through named ligaments or omenta.

VISCERAL RELATIONS

Anterior relations

Right part of anterior surface: related to gastric impression of left lobe of liver and near pylorus to quadrate lobe of liver.

Left half: related to diaphragm and rib cage.

Lower part: related to anterior abdominal wall.

Gastric triangle: part in contact with anterior abdominal wall; useful in gastrostomy.

Posterior relations: stomach bed

The posterior surface is related to diaphragm, left kidney, left suprarenal gland, pancreas, transverse mesocolon, left colic flexure and splenic artery.

Mnemonic: Dr S3 Kills Patients Mercilessly = Diaphragm, Splenic artery, Suprarenal gland, Splenic flexure, Kidney, Pancreas, Mesocolon.

Fig. 7.6 - Anterior relations of the stomach

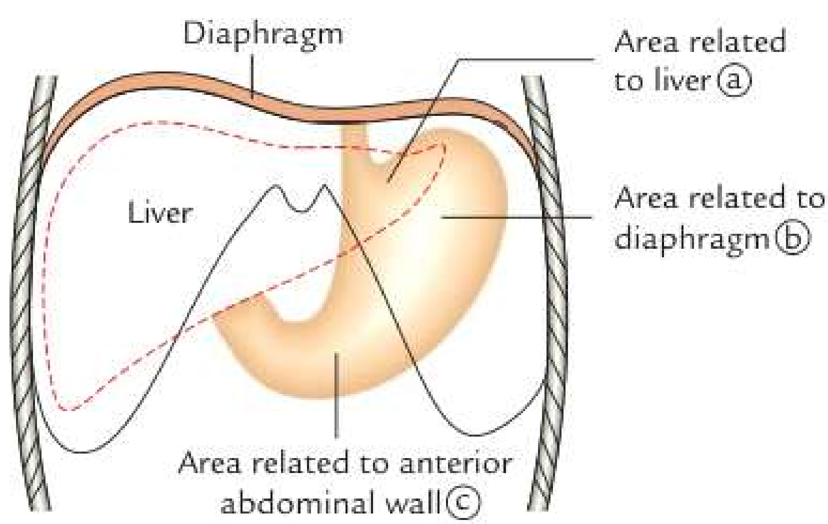
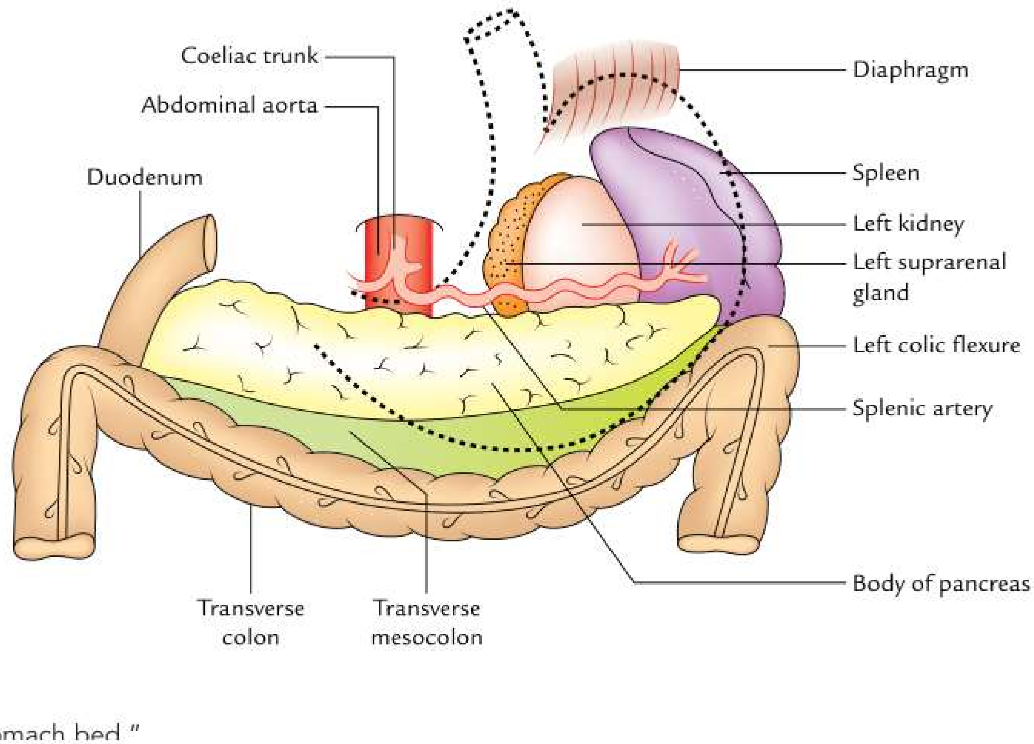


Fig. 7.6 Anterior relations of the stomach.

The anterior surface is divided into areas related to liver, diaphragm and anterior abdominal wall. The anterior abdominal wall contact area is clinically important because this is the approach used for gastrostomy when feeding access is required.

Fig. 7.7 - The stomach bed



The stomach bed is a collection of structures posterior to the stomach. The lesser sac separates the stomach from most of these structures. This makes posterior relations very important in posterior gastric ulcer, pancreatic pathology and spread of inflammation.

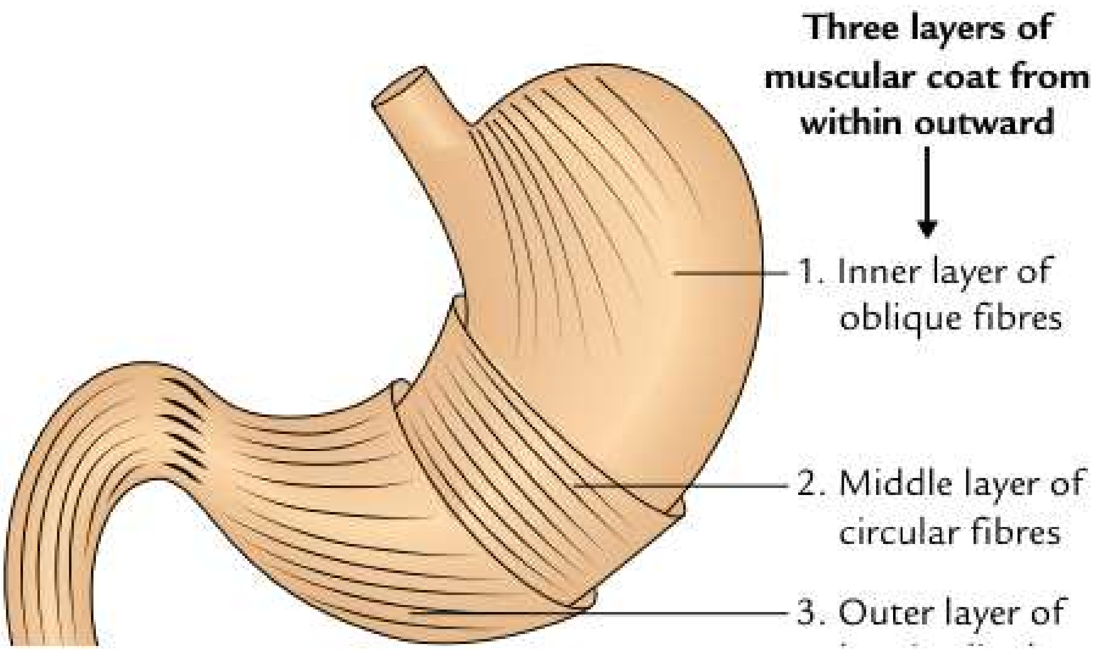
MICROSCOPIC STRUCTURE AND INTERIOR

Wall of the stomach

From outside inward, the wall has four coats: serous, muscular, submucous and mucous. The muscular coat is especially important because the stomach has three layers of smooth muscle rather than the usual two layers of most gut segments.

- Serous coat: peritoneum.
- Muscular coat: outer longitudinal, middle circular and inner oblique layers.
- Submucous coat: loose areolar tissue.
- Mucous coat: thick, soft, velvety, with gastric folds/rugae and gastric glands.

Fig. 7.8 - Three layers of muscular coat

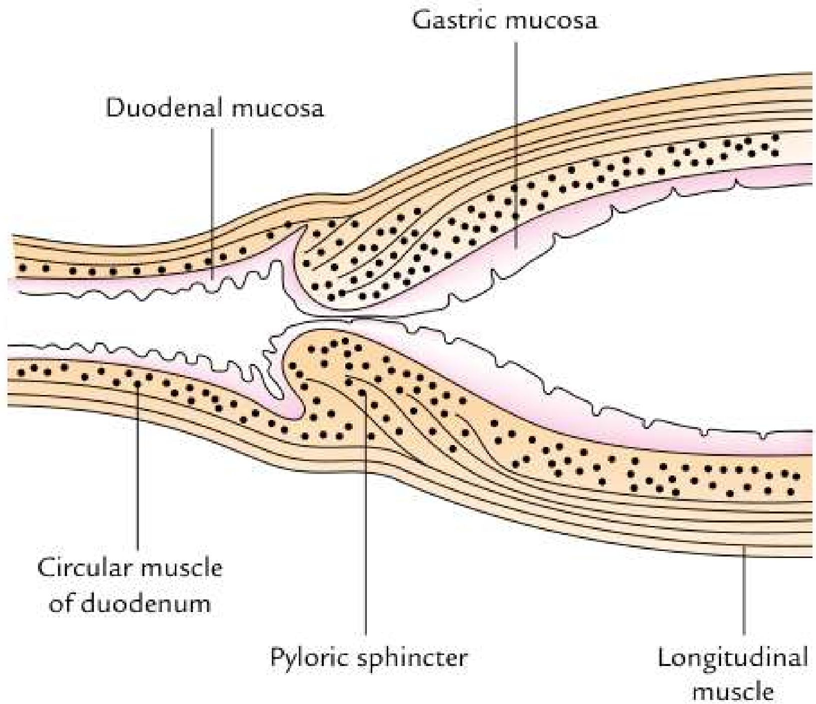


The extra inner oblique layer helps the stomach churn and mix food efficiently. Near the pylorus, circular muscle thickens to form the pyloric sphincter; this anatomical sphincter controls the passage of chyme into the duodenum.

How to explain the muscle coat in an exam

Write the layers from outside inward as longitudinal, circular and oblique. The oblique fibres are characteristic of the stomach and help powerful churning movements. The circular layer becomes thick at the pylorus to form the anatomical pyloric sphincter, while the longitudinal fibres assist the sphincter by turning inward near the pyloric end.

Fig. 7.9 - Pyloroduodenal junction and pyloric sphincter

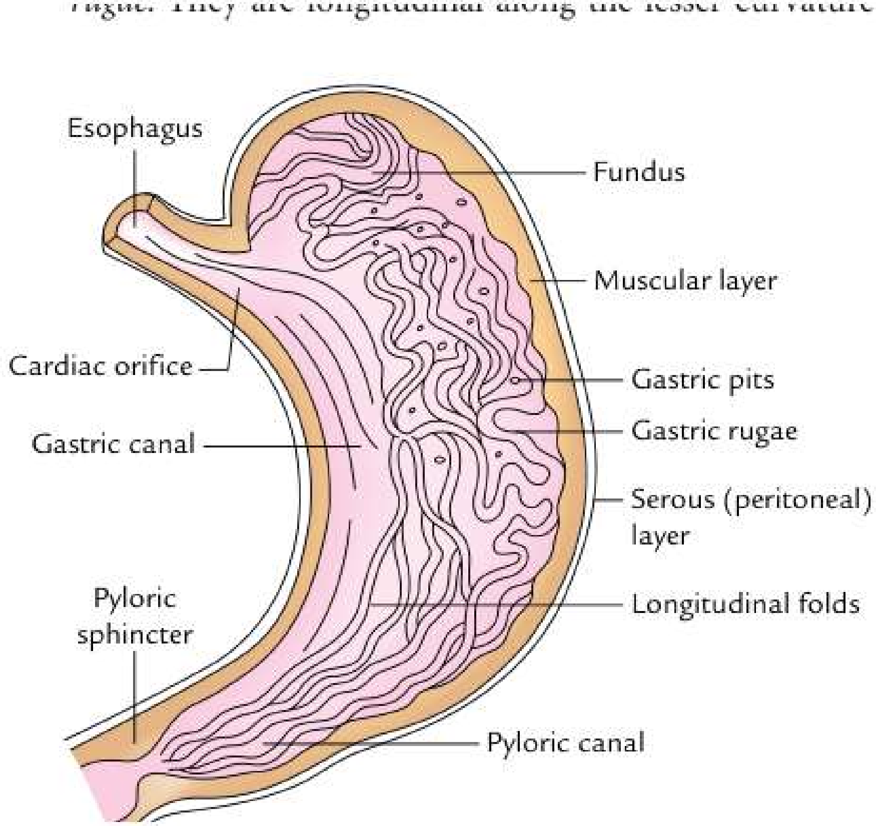


The pyloric sphincter is mainly due to thickening of the circular muscle coat, assisted by longitudinal fibres. This figure explains why pyloric obstruction or spasm affects gastric emptying and may lead to vomiting and gastric dilatation.

Pyloric region - why it is important

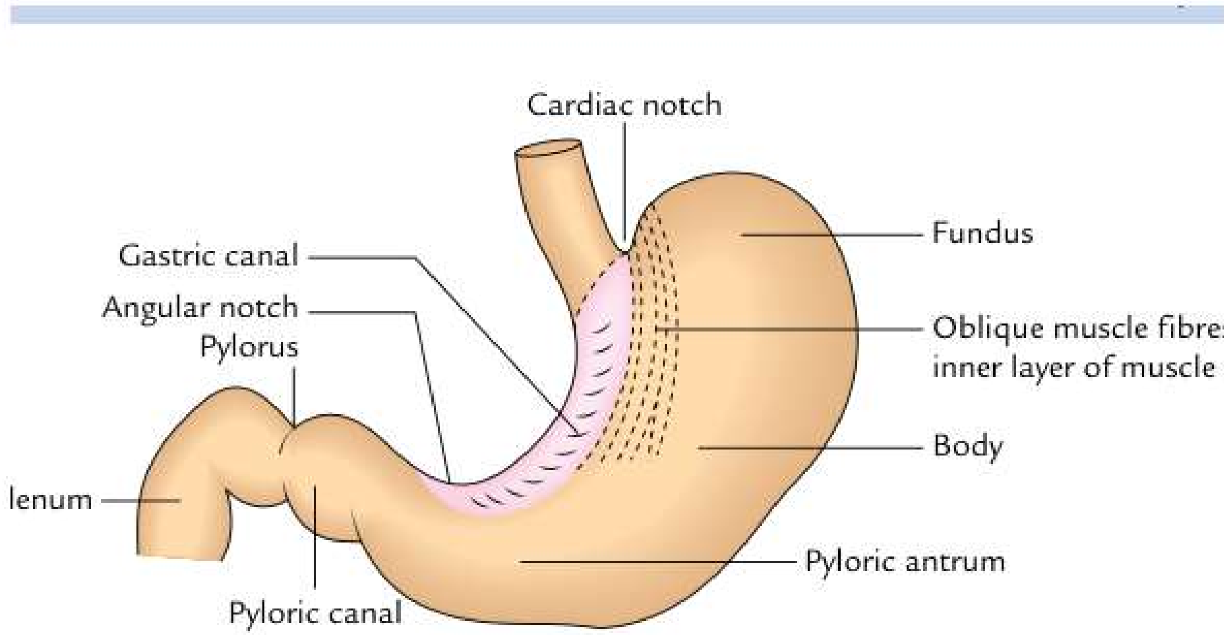
The pylorus is a true anatomical sphincter and acts as a gatekeeper between stomach and duodenum. It prevents rapid emptying of large food particles and regulates the rate at which acidic chyme enters the duodenum. Clinically, pyloric stenosis or obstruction produces visible gastric peristalsis, vomiting and delayed gastric emptying.

Fig. 7.10 - Interior of the stomach



The inner mucosa forms gastric rugae, which flatten when the stomach distends. The gastric canal is a temporary channel along the lesser curvature that may allow swallowed liquids to pass rapidly towards the pyloric region before spreading to the rest of the stomach.

Fig. 7.11 - Gastric canal



The gastric canal is located along the lesser curvature. Because liquids and irritants may move rapidly along this channel, the lesser curvature is exposed to greater mucosal insult and is a common site for gastric ulceration.

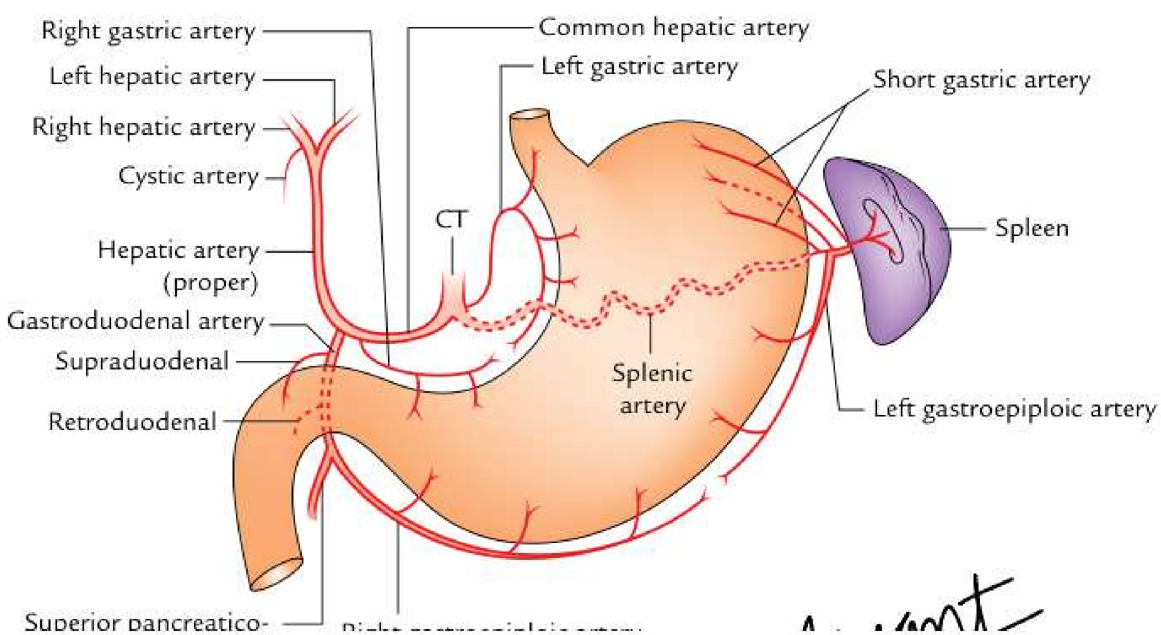
ARTERIAL AND VENOUS DRAINAGE

Arterial supply

The stomach has a rich arterial supply derived from the coeliac trunk and its branches. The vessels run mainly along the lesser and greater curvatures and form anastomotic channels.

- Left gastric artery: direct branch of coeliac trunk.
- Right gastric artery: branch of common hepatic artery.
- Left gastroepiploic artery: branch of splenic artery.
- Right gastroepiploic artery: branch of gastroduodenal artery.
- Short gastric arteries: 5-7 branches of splenic artery.

Fig. 7.12 - Arteries of the stomach



The lesser curvature receives left and right gastric arteries. The greater curvature receives left and right gastroepiploic arteries. The fundus receives short gastric arteries from the splenic artery; these are important during splenic surgery because they can be injured when handling the gastrosplenic ligament.

Venous drainage

Gastric veins correspond to the arteries and drain directly or indirectly into the portal vein.

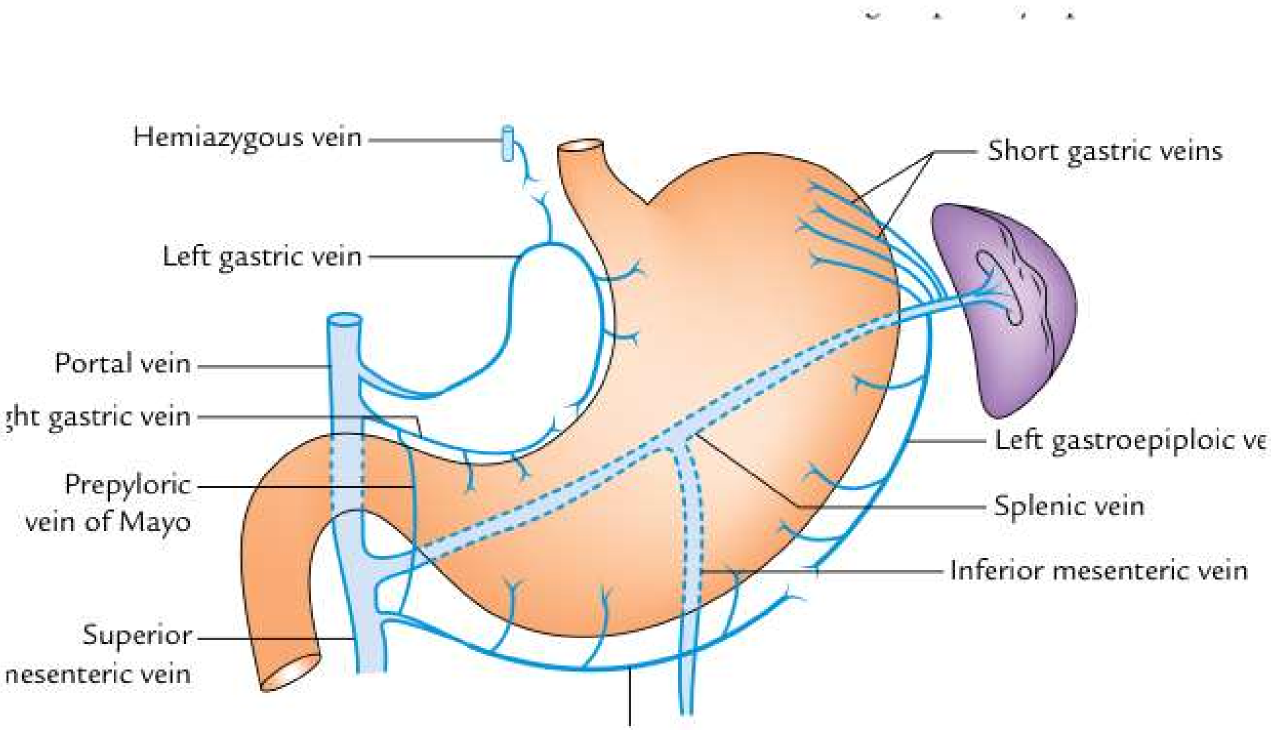
- Left and right gastric veins drain directly into portal vein.
- Left gastroepiploic and short gastric veins drain into splenic vein.
- Right gastroepiploic vein drains into superior mesenteric vein.

Clinical link

The lower end of esophagus drains partly into portal and partly into systemic circulation, creating a portocaval anastomosis.

- In portal hypertension, dilated esophageal veins may form varices.
- Rupture can cause hematemesis.

Fig. 7.13 - Venous drainage of the stomach



Follow the blue veins towards the portal system: gastric veins enter portal vein, left gastroepiploic/short gastric veins enter splenic vein and right gastroepiploic enters superior mesenteric vein. This diagram is a compact map of venous return from the stomach.

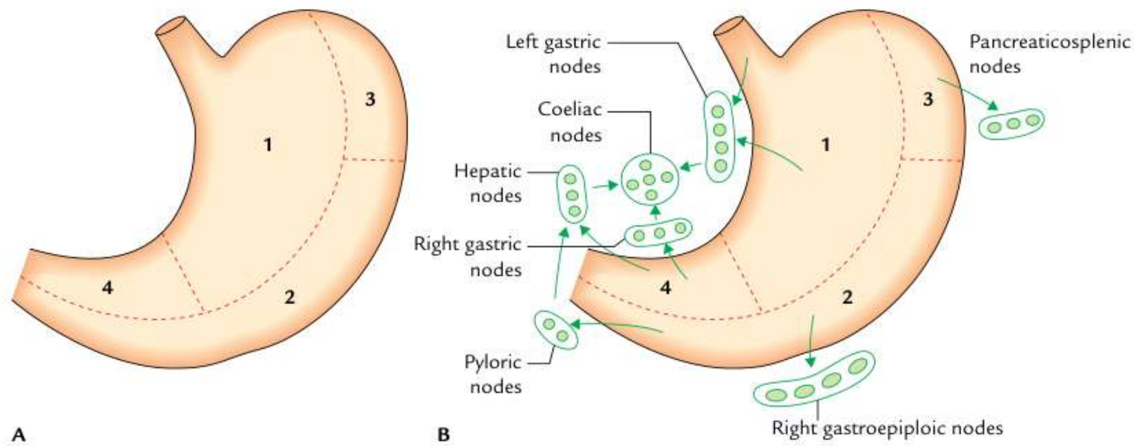
LYMPHATIC DRAINAGE

Lymphatic territories

For descriptive lymph drainage, the stomach is divided into four territories. This arrangement is clinically essential because gastric carcinoma spreads through lymph vessels and may first present by nodal enlargement.

- Area 1: largest area along lesser curvature -> left gastric nodes -> coeliac nodes.
- Area 2: pyloric antrum/canal along greater curvature -> right gastroepiploic and pyloric nodes.
- Area 3: pancreaticosplenic area -> pancreaticosplenic nodes along splenic artery.
- Area 4: pyloric antrum/canal along lesser curvature -> right gastric and hepatic nodes.
- Efferents from all groups finally pass to coeliac nodes, then intestinal lymph trunk and cisterna chyli.

Fig. 7.14 - Lymphatic drainage of the stomach



ig. 7.14 Lymphatic drainage of the stomach: A, lymphatic territories; B, lymph node groups draining the lymphat

Diagram A shows four lymphatic territories; diagram B shows nodal groups. The pyloric antrum along the greater curvature is especially important because gastric cancer commonly occurs here. Spread through lymphatics may reach the left supraclavicular node, producing Virchow node and Troisier sign.

Applied anatomy - gastric carcinoma

Gastric carcinoma commonly occurs in the region of pyloric antrum along the greater curvature. It spreads through lymph vessels to the left supraclavicular lymph nodes. An enlarged left supraclavicular node is called Virchow node, and its presence as a first sign of gastric cancer is called Troisier sign.

NERVE SUPPLY

Sympathetic supply

Derived from T6-T10 spinal segments through greater splanchnic nerves and coeliac/hepatic plexuses.

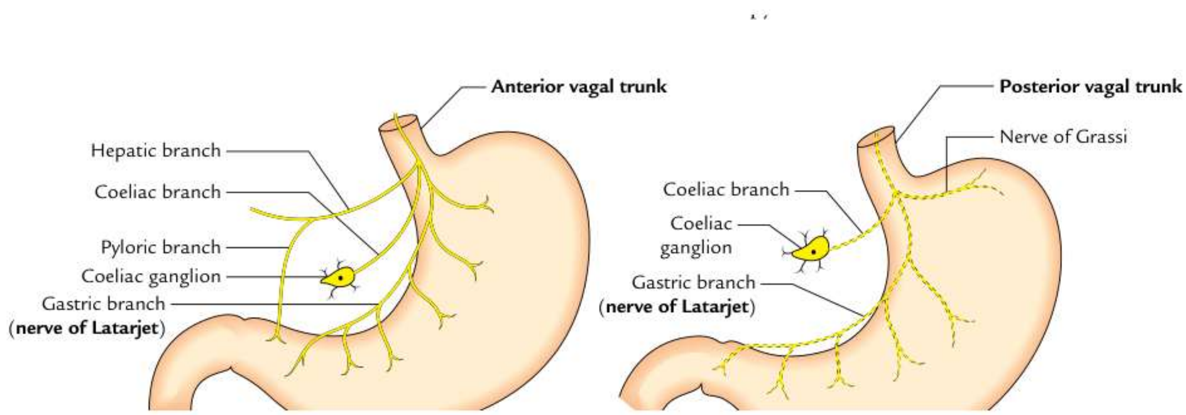
- Fibres reach stomach along arteries.
- Functions: vasomotor, motor to pyloric sphincter, inhibitory to remaining gastric musculature.
- Main pathway for pain; gastric pain is referred to epigastrium.

Parasympathetic supply

Derived from vagus nerves.

- Anterior vagal trunk: mainly from left vagus; gives hepatic, coeliac and gastric branches.
- Posterior vagal trunk: mainly from right vagus; gives coeliac branch, nerve of Grassi and posterior gastric branch.
- Nerves of Latarjet supply acid/pepsin secreting areas and gastric branches.

Fig. 7.15 - Parasympathetic innervation of the stomach



The anterior vagal trunk supplies the anterior surface and gives hepatic, coeliac and gastric branches. The posterior vagal trunk supplies the posterior surface and gives branches including the nerve of Grassi. These branches are central to understanding vagotomy procedures for peptic ulcer disease.

Vagotomy - applied summary

- Truncal vagotomy: both vagal trunks divided at lower end of esophagus; decreases acid but affects gastric emptying.
- Selective vagotomy: hepatic and coeliac branches preserved; gastric branches/nerve of Latarjet cut; pyloric antrum may be denervated.
- Highly selective vagotomy: only parietal-cell area is denervated; gastric emptying is better preserved and it is considered more physiological.

APPLIED ANATOMY AND EXAM POINTS

Common applied points

Gastric ulcer often occurs along lesser curvature because the gastric canal exposes this region to swallowed irritants and acid.

- Pyloric obstruction may produce vomiting and gastric dilatation.
- Gastric pain is usually referred to epigastrium due to T6-T10 segmental supply.
- Gastric carcinoma may spread to Virchow node.

Must-remember facts

Widest and most distensible part of alimentary canal: stomach.

- Usual shape: J-shaped.
- Adult capacity: 1500-2000 ml.
- Anatomical sphincter: pyloric sphincter.
- Posterior relations: stomach bed.
- Final lymph drainage: coeliac nodes.

Question	High-yield answer
Most fixed end	Cardiac end.
More mobile end	Pyloric end.
Fundus level	Usually reaches left 5th intercostal space just below nipple.
Structures forming stomach bed	Diaphragm, left kidney, left suprarenal gland, pancreas, transverse mesocolon, left colic flexure, splenic artery.
Main arteries	Left/right gastric, left/right gastroepiploic, short gastric arteries.

Pain referral	Epigastric region through T6-T10 segments.

Quick revision paragraph

The stomach is a J-shaped, highly distensible organ located mainly in the left hypochondrium and epigastrium. It has two ends, two curvatures, two surfaces and four parts. Lesser curvature carries the angular notch and lesser omentum; greater curvature carries the greater omentum, gastrosplenic and gastrophrenic ligaments. Posteriorly, the stomach is related to the stomach bed through the lesser sac. Its blood supply comes from coeliac trunk branches, venous drainage enters the portal system, lymph finally drains to coeliac nodes and nerve supply comes from vagus and T6-T10 sympathetic fibres.

Last-minute exam checklist

- Draw and label the J-shaped stomach with fundus, body, pyloric antrum, pyloric canal and pylorus.
- Mention both orifices and their vertebral/surface levels.
- Write anterior relations separately from posterior relations/stomach bed.
- In blood supply, group arteries along lesser curvature, greater curvature and fundus.
- In lymph drainage, remember final drainage to coeliac nodes and clinical link with Virchow node.
- In nerve supply, mention vagal trunks, nerves of Latarjet, T6-T10 sympathetic fibres and epigastric referred pain.